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Assessment of Critical Functions in Battery Production Lines for Predictive Maintenance

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Abstract

In an increasingly automated and data-driven manufacturing environment, the utilization of real-time data for predictive maintenance plays a crucial role in minimizing downtime and improving reliability. This thesis focuses on identifying and assessing critical functions in the battery module production lines at Scania's battery plant in Södertälje, Sweden. The goal is to support smart maintenance by evaluating where the pain points exist in the battery production lines and ensuring that these areas are effectively monitored.

A condition-based risk ranking model was developed using three key parameters: failure visibility, frequency, and downtime. Data for this analysis was extracted from Scania's existing process monitoring system (PROMO) and the Computerized Maintenance Management System (CMMS) - IBM Maximo. To structure and understand the production process flow as functions, the Function Analysis System Technique (FAST) was used at both the machine and overview levels. This helped create a logical mapping of how, why, and when each function occurs.

Based on the functional risk assessment, this work also evaluates the sufficiency of the current sensor and vision system coverage. For functions with inadequate monitoring, sensor or vision system upgrades are proposed, supported by both functional risk justification and historical qualitative work orders from Maximo. These findings will be used to support dysfunctional analysis for future research. Additionally, a conceptual framework for integrating Machine Learning (ML) is proposed to support maintenance teams by providing data-driven recommendations for incoming work orders.

The outcome of this thesis provides a pathway toward smarter, more proactive maintenance planning that is grounded in operational data and human reasoning, a practical step toward more resilient and efficient EV battery production lines.

Keywords

Battery production; FAST; Condition-based risk ranking; CMMS (IBM Maximo); Process monitoring; Sensor adequacy; Predictive maintenance; Machine learning; EV manufacturing; Smart maintenance

Sammanfattning

I en allt mer automatiserad och datadriven tillverkningsmiljö spelar användningen av realtidsdata för prediktivt underhåll en avgörande roll för att minimera stillestånd och förbättra tillförlitligheten. Denna uppsats fokuserar på att identifiera och bedöma kritiska funktioner i batterimodulens produktionslinjer vid Scanias batterifabrik i Södertälje, Sverige. Målet är att stödja smart underhåll genom att utvärdera var flaskhalsar uppstår i produktionslinjerna och säkerställa att dessa områden övervakas på ett effektivt sätt.

En tillståndsbaserad riskrankningsmodell utvecklades med tre nyckelparametrar: felsynlighet, frekvens och stilleståndstid. Data för denna analys hämtades från Scanias befintliga processövervakningssystem (PROMO) och det datoriserade underhållssystemet (CMMS) – IBM Maximo. För att strukturera och förstå produktionsflödet som funktioner användes Function Analysis System Technique (FAST) på både maskin- och översiktsnivå. Detta hjälpte till att skapa en logisk kartläggning av hur, varför och när varje funktion uppträder.

Baserat på den funktionella riskbedömningen utvärderas även täckningen och tillräckligheten hos nuvarande sensor- och visionssystem. För funktioner med otillräcklig övervakning föreslås uppgraderingar av sensorer eller visionssystem, med stöd av både funktionell riskanalys och historiska arbetsordrar från Maximo. Dessa resultat kan användas för att stödja dysfunktionsanalys i framtida forskning. Dessutom föreslås ett konceptuellt ramverk för att integrera Maskininlärning (ML) som ett stöd för underhållsteam genom att ge datadrivna rekommendationer för inkommande arbetsordrar.

Resultatet av denna uppsats ger en väg mot smartare och mer proaktiv underhållsplanering som är baserad på operativ data och mänsklig analys – ett praktiskt steg mot mer robusta och effektiva produktionslinjer för elbatterier.

Nyckelord

Batteriproduktion; FAST; Tillståndsbaserad riskrankning; CMMS (IBM Maximo); Processövervakning; Sensortäckning; Prediktivt underhåll; Maskininlärning; Elfordonstillverkning; Smart underhåll

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1. Introduction

1.1 Background and Motivation

Scania's modern battery production lines are at the heart of the electrification and digitalization process. These systems must maintain high throughput, traceability, and reliability while managing increasingly complex, robotic, automated processes. In such conditions, unexpected equipment failures or unplanned downtime can lead to significant production losses, especially in new factories where operational maturity is still evolving.

At Scania's battery factory, which is a state-of-the-art and heavily automated facility, ensuring process stability and equipment uptime is crucial. As production volumes increase and systems become more interoperable and interconnected, reactive (Immediate Corrective Maintenance and Deferred Corrective Maintenance) or time-based maintenance (Preventive Maintenance) strategies are no longer sufficient. Instead, there is a growing need to include predictive maintenance (PdM), which highly relies on operational real-time data, smart diagnostics, and taking actions before breakdowns occur.

To successfully develop and deploy a predictive maintenance strategy in such an environment, the first step is to identify which functions, such as sensors, vision systems, actuators, and motors, within the production line are most critical. Without understanding what needs to be identified, maintenance resources may be misallocated, resulting in either over-maintenance or neglected potential risk areas. This thesis addresses that gap by applying a function-based criticality assessment methodology that utilizes operational logic, historical maintenance, process data, and sensor feedback to guide maintenance planning in a data-driven, actionable way.

1.2 Problem Statement

The challenge lies not only in detecting faults but in understanding where those faults will have the highest impact on the production line. At Scania's high-speed battery module production line, hundreds of interlinked functions exist across mechanical, electrical, automation, and software systems. Not all failures are equal, but some can stop production entirely, while others may bring down machine performance slowly over time.

Current maintenance and process monitoring systems log alarms, work orders, and monitor real-time signals, but without a structured analysis of functional importance and monitoring adequacy, maintenance decisions remain reactive. So, a systematic approach is required to:

- Map the production process by its functions.
- Quantify risk using real-time and historical data.
- Evaluate the adequacy of existing monitoring systems.
- Propose and reason improvements based on criticality.

1.3 Research Questions

This thesis focuses on answering the following research questions:

1. How can we use a function-based approach with real-time signals and historical maintenance data?

2. How can this approach be used to enhance maintenance decision-making at the battery factory?

1.4 Aim and Objectives

The aim of this thesis is to assess critical machine-level functions in the battery module production line in order to support the development of a predictive maintenance strategy. To achieve this aim, several objectives were defined. The first was to map both overview- and machine-level functions using the Function Analysis System Technique (FAST). The second was to rank these functions by criticality through the application of a condition-based risk ranking model that incorporates failure frequency, downtime, and failure visibility. A third objective was to evaluate the current monitoring systems, including sensors, vision systems, motors, and actuators, and to identify potential monitoring gaps.

Where functions were identified as high risk and detection was insufficient, recommendations were made for signal improvements or the introduction of additional monitoring systems. Finally, the thesis proposes an AI/ML-based framework that can support maintenance decision-making by learning from historical CMMS data, specifically the work orders recorded in Maximo.

1.5 Scope and Limitations

This study was carried out within the context of Scania's battery module production line. The scope of the analysis was limited to machine-level functions, without extending to sensor-level assessment. The research was further bound by the availability of data, relying on maintenance records from the CMMS (Maximo) system and process data captured by the process monitoring platform. Machine-level functions (not sensor-level granularity).

In terms of predictive modelling, the work concentrated on the conceptual design of an AI/ML-based framework, without attempting full model deployment. Certain information, such as process station details, supplier data, design specifications, or control logic, was excluded from the analysis to comply with confidentiality agreements. These boundaries defined the extent of the research while ensuring that the findings remained relevant to the operational needs of the production line.

1.6 Applicability to Sustainability and Smart Factory

The work presented in this thesis also aligns with broader goals of sustainability and smart manufacturing. By improving equipment uptime and reducing unnecessary maintenance, the research contributes directly to several of the United Nations Sustainable Development Goals (SDGs). It supports SDG 9: Industry, Innovation, and Infrastructure by strengthening the reliability of data-driven manufacturing systems. It contributes to SDG 12: Responsible Consumption and Production by helping to reduce part waste, labour costs, and excess energy consumption associated with unplanned stoppages. Finally, it indirectly supports SDG 13: Climate Action by improving the efficiency of electric vehicle battery production processes, thereby contributing to the reduction of CO_2 emissions across the supply chain.

2 Literature Review

2.1 Predictive Maintenance in Industry 4.0

Modern Scania battery production systems are not just mechanical and manually operated; they are cyber-physical networks where sensors, vision systems, automation, and several data platforms work together. This is the core materialisation of Industry 4.0 concepts: connecting real-time shop-floor production to decision-making platforms. Predictive maintenance (PdM) is a key player, aiming to identify equipment failures before they cause production downtime.

Several studies consistently report that PdM reduces downtime and improves resource and material utility, but they also point out a major challenge: it only works when it is completely connected to the external environments that influence the plant's production and is supported by reliable data sources [1]. For example, in highly automated and critical environments such as battery module assembly, even small stoppages can be responsible for significant production losses, making the ability to anticipate failures far more valuable than in less tightly coupled processes [2].

In battery manufacturing specifically, PdM is increasingly recognised as a strategic motivator for sustainability. This is not only because it reduces waste and energy loss, but also because it ensures the consistent throughput needed to meet EV demand [3].

2.2 Maintenance Strategies – Moving from Time-Based to Evidence-Based

Maintenance thinking and strategies have changed dramatically in recent decades. Early practices were dominated by reactive maintenance, where intervention took place only after a failure had already occurred. Within this category, Immediate Corrective Maintenance (ICM) refers to actions taken right after a breakdown, focused on quickly restoring the system to working condition. These repairs are usually unplanned and may cause sudden production stoppages, but they are unavoidable when failures directly interrupt operations or create safety risks.

In contrast, Deferred Corrective Maintenance (DCM) is applied when a fault is detected but does not immediately stop the process or compromise safety. In such cases, maintenance is scheduled for a later time, often during a planned maintenance period, when the necessary resources such as spare parts or staff are available.

Traditional preventive maintenance (PM) schedules corrective actions based on time or usage, regardless of the actual condition of the asset. While this can prevent some failures, it often leads to unnecessary work and premature component replacement.

Condition-based maintenance (CBM) improves on this by using sensor threshold limits or inspections to decide when intervention is needed [4]. CBM is more efficient than PM, but it still reacts to signs of a decrease in asset value; it does not predict when that deterioration might go beyond the threshold and reach a critical point.

This is where predictive maintenance comes in. By combining real-time machine and monitoring system data, historical records, and analytical models, PdM estimates the remaining useful life of components and recommends maintenance just before failure is likely to occur.

The general agreement in literature is that no single strategy fits all assets; a hybrid model integrating PM, CBM, and PdM gives the most resilient system.

The battery module line already uses several reactive, preventive, and condition-based measures. Identifying critical functions and ranking them by risk is the bridge that makes predictive strategies targeted and effective, rather than generic.

2.3 CMMS (Maximo) and Smart Maintenance Platforms

A Computerized Maintenance Management System (CMMS) like IBM Maximo is more than a planning and scheduling tool. It is the memory of the maintenance activities done in the battery factory. It records work orders, downtime data, parts usage, and most importantly, technicians' qualitative notes.

Several studies show that CMMS data is underused, especially the work orders sections where technicians record observations, symptoms, and informal diagnostic measures [5]. This is a missed opportunity because these records often explain *why* something failed, not just *that* it failed. Analysing this data can reveal recurring failure modes, highlight technical training supports, and even reveal design limitations [6].

At the Scania battery plant, Maximo is one of the main maintenance data sources. Utilising both quantitative data (e.g., downtime) and qualitative descriptions of the maintenance issue, combined with process monitoring data from Scania's process monitoring platform, creates a much richer picture of functional risk than either system could provide alone.

2.4 Function Analysis (FAST)

The FAST methodology comes from value engineering, but its logic and mapping functions in a *"How?"* and *"Why?"* sequence make it particularly useful in maintenance planning [7]. By reducing each part of a system to a verb–noun function, FAST creates a common language that engineers, operators, and planners can share.

In practice, FAST diagrams make it easier to spot dependencies. If one function fails, it can be traced to a higher-level dependent system, and also to which lower-level system might also be impacted. This is why it is effective in identifying critical functions, those without which the entire system goal cannot be achieved [8].

In this research, FAST is the foundation. It is applied at both the overview and machine levels in the battery module line. Once those functions were mapped, employing condition-based risk ranking essentially turned a static diagram into a data-driven decision-support tool.

2.5 Condition-Based Risk Ranking

Knowing which functions exist is only the first step. Knowing which are the riskiest is crucial for implementing predictive maintenance strategies. Risk-based maintenance uses a combination of likelihood and consequence to prioritise work. Standards such as IEC 60812 (2018) and ISO/IEC 31010 (2019) outline formal approaches to scoring these factors.

The analysis relied on three key parameters. Frequency of failure, extracted from Maximo work orders, was used as an indicator of likelihood. Downtime impact, based on Maximo records and maintenance logs, reflected the consequence of a failure. Finally, failure visibility, derived from sensor coverage and alarms in the process monitoring platform, represented detectability.

By quantifying these three parameters and calculating a combined score, a ranked list of critical functions was created. This ranking was then used to analyse sensor adequacy, revealing where existing monitoring was sufficient and where additional monitoring systems (sensors, vision systems) would be justified.

3 Methodology

3.1 Overview of the Battery Production Line

The battery production line under study is a highly automated, complex system designed to assemble modules for electric vehicle battery packs. The line integrates robotics, material handling systems, and in-line quality assurance to achieve high throughput with consistent precision. While each stage performs a distinct role in the assembly sequence, the production flow is continuous and interlinked, meaning any stoppage in one area quickly impacts the entire system.

To preserve confidentiality, individual stations are referred to here as Machine A, Machine B, and Machine C, representing different stages of the production process. Each of these machines combines several key supporting elements.

Industrial robots, both 3-axis and 5-axis, are employed for handling, positioning, and assembly tasks. Conveyors and Automated Guided Vehicles (AGVs) transport materials and components between stations with minimal human intervention. Integrated testing equipment performs electrical, mechanical, or positional checks to ensure quality before progression. A variety of sensors, such as contact, non-contact, and environmental, along with monitoring parameters such as alignment, presence, temperature, or force during crucial operations. Vision systems support surface inspection, part verification, and positional feedback.

Control and monitoring are achieved through Programmable Logic Controllers (PLC) at each station, which are interfaced with Supervisory Control and Data Acquisition (SCADA) systems. Human-Machine Interfaces (HMI) enable operators to interact with station control logic and carry out manual overrides when required. At a higher level, the Manufacturing Execution System (MES) coordinates production schedules, tracks work-in-progress, and links operational data to broader planning systems. In parallel, the internal Process Monitoring System records both real-time and historical process data, ensuring traceability and enabling detailed analysis.

The integration of these systems ensures that production is both data-rich and traceable, providing the foundation for predictive maintenance (PdM) strategies. Studies on similar high-throughput manufacturing environments highlight that data accessibility and interoperability between shop-floor devices and information systems are prerequisites for effective PdM [9].

From a maintenance perspective, the line presents several operational challenges. Interdependencies between machines mean that failures are rarely isolated, as a stoppage in Machine A can quickly cause downstream idle time or an increase in buffer in Machines B and C. The diversity of equipment types, ranging from robotics to conveyors and testers, requires equally diverse maintenance strategies and skillsets. Furthermore, the high costs associated with changeovers make unplanned downtime particularly expensive.

Because of these factors, understanding the functional role of each machine and the criticality of its sub-functions is vital before implementing predictive models. This justifies the choice of function analysis as the starting point for this thesis, allowing a logical, system-wide understanding of how each function supports the overall production goal before applying condition-based risk ranking [10].

3.2 Functional Breakdown using FAST

Before analysing equipment performance or predicting failures, it is essential to understand what functions each part of the system is meant to fulfil. Without this clarity, maintenance decisions risk being reactive, fragmented, and disconnected from the actual purpose of the equipment. The Function Analysis System Technique (FAST) provides a structured way to achieve this understanding.

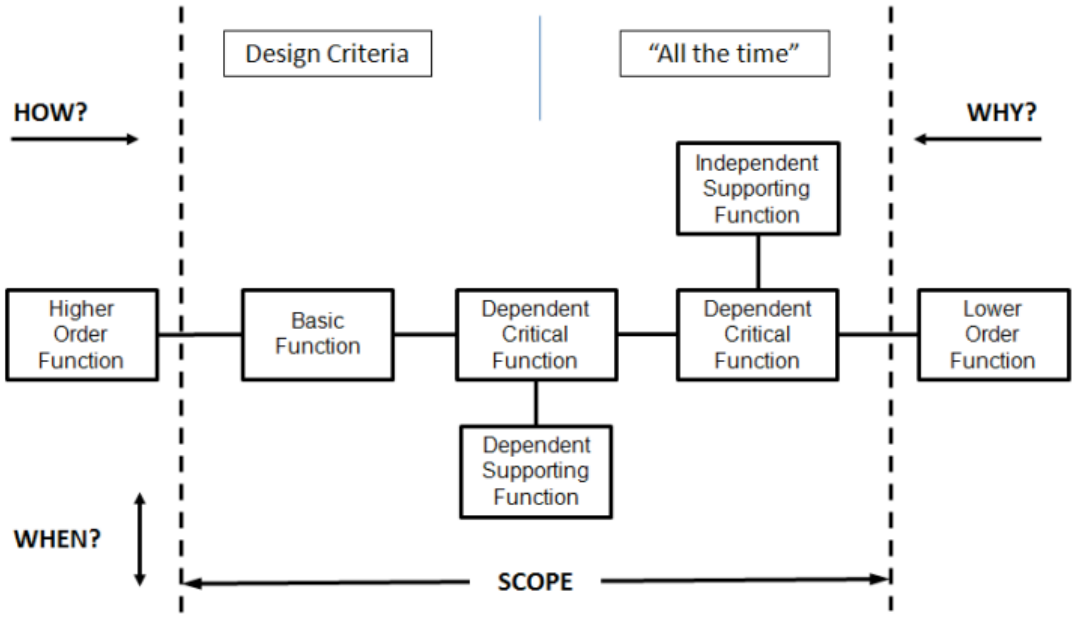


Figure 1: Basic layout of FAST diagram for Function breakdown [8]

FAST was originally developed in value engineering to help teams map the relationship between functions and the means of achieving them [7]. In manufacturing maintenance, the method has gained attention because it cuts through design complexity and focuses on why a function exists, how it is achieved, and what it enables [8]. This makes it well-suited for automated production lines where multiple machines, sensors, and control systems interact to achieve a common production goal.

3.2.1 Why FAST was Chosen for this Thesis

In the battery module production line under study, there are multiple machines (Machine A, Machine B, Machine C), each containing several sub-systems. These include robots for material handling, conveyors and AGVs for part transfer, in-line testers for quality assurance, sensor networks for process monitoring, and vision systems for inspection.

While each sub-system has its own maintenance requirements, their functional relationships determine the real impact of a failure. For example, if a vision system in Machine B fails, the function it supports (e.g., verifying component placement) might have downstream effects that impact Machine C's ability to operate. FAST captures these dependencies in a function logic diagram, which later becomes the foundation for risk scoring.

By asking structured questions such as *"How is this function achieved?"* and *"Why is this function necessary?"*, FAST builds a logical chain that connects individual tasks to the overall production objective [11]. This approach is especially useful in situations where documentation is incomplete or outdated, as is often the case in newly established production lines. It also supports knowledge-sharing when expertise is distributed across different teams, such as process engineering, production, IT, and maintenance. Furthermore, in highly interconnected systems where failure effects are not always intuitive, FAST provides clarity by mapping how each function contributes to the wider process.

3.2.2 How FAST was Applied in this Thesis

The FAST methodology was applied at two levels: the overview level and the machine level.

1. **Overview level**

   At the highest level, FAST was used to identify the core functional flow of the production line, represented here by Machines A, B, and C. An example of this mapping is the sequence "Assemble module" leading to "Position cell" and further to "Pick cell." Each of these high-level functions is critical, since the failure of any one would prevent completion of the final product. The overview level, therefore, established the backbone of the process, onto which more detailed functional dependencies could later be mapped.

2. **Machine level**

   At the next level, each machine was broken down into its internal sub-functions, which revealed the detailed interactions between components. For example, "Pick cell" could be further mapped into "Actuate sensor A" and "Acknowledge gripper pressure." This sequence describes the actions involved when the system receives a command to pick a component. A robot arm positions itself over the part, a sensor detects readiness and actuates a signal to the control system, and finally, the system acknowledges the gripper's pressure feedback to confirm that the component has been secured before proceeding.

This level of mapping is especially valuable because it highlights the supporting functions, such as sensors, actuators, and control feedback, that may not appear in the overview level but are essential for the process to operate reliably. Failures at this level may occur more visibly and deeply than at the overview level, but their effects can still accumulate through the production line if left unaddressed [11].

3. **Sensor level**

   The FAST approach can be extended to an even more granular sensor level, where each function is linked directly to PLC logic, signal flow, and physical sensing elements. At this level, it would be possible to map the cause-and-effect chain of individual sensor

signals, actuator responses, and subsequent acknowledgements. However, given the scope and time constraints of this master's thesis as well as confidentiality limitations, this level of mapping was not conducted. The focus remained on the overview and machine levels, which still capture the functional structure needed for criticality assessment.

3.2.3 Linking FAST Functions to Maintenance Data

Instead of creating unique function IDs for cross-referencing each overview and machine-level function, which would be a logical approach in large-scale systems and is suggested for future work, this thesis directly linked FAST functions to maintenance work orders. This involved reviewing work orders in the CMMS (Maximo), with close attention to the free-text problem descriptions. The reported sensor or actuator was identified, without disclosing exact tag names, and then manually verified against the FAST diagram to determine which overview- and machine-level functions it supported. Each work order was subsequently mapped to its corresponding functions at both levels, ensuring consistency for later risk ranking.

Although this manual linking process was time-consuming, it ensured that each maintenance record could be meaningfully associated with a functional role in the production line. This mapping later served as the basis for condition-based risk scoring, where the frequency, downtime, and visibility of failures were quantified.

Advantages of Using FAST in Predictive Maintenance Planning

The application of FAST offers several advantages for predictive maintenance planning. One of the most significant is improved communication, as the method creates a common understanding across process engineering, IT, and maintenance teams. It also provides dependency awareness by making interconnections transparent, reducing the likelihood of overlooking indirect impacts of a failure.

In addition, it establishes a solid foundation for risk analysis, since once functions are identified, they can be evaluated for criticality using historical downtime, failure frequency, and visibility metrics. FAST also supports the integration of smart automation, as function identifiers can serve as key parameters for linking real-time signals to predictive models.

These advantages are supported by studies showing that functional modelling approaches can improve asset criticality analysis and optimise monitoring strategies in automated environments.

3.3 Data Sources: Process Monitoring, CMMS, and Maintenance Logs

A functional breakdown using the FAST methodology provides the structural map of how a production line operates, but on its own, it does not tell us how often a function fails, how long it takes to recover, or how visible such failures are during operation. To gain these insights, the FAST model needs to be linked to real operational data sources. For this research, two key categories of data were used:

1. Computerised Maintenance Management System (CMMS), which contains detailed records of reported failures, maintenance actions, and component-level issues.

2. Process Monitoring Data, which captures both real-time and historical machine and sensor performance parameters.

These sources serve complementary purposes. The process monitoring system provides quantitative process behaviour, while CMMS data captures qualitative and event-based maintenance information. Together, they provide a holistic picture of production system health and form the basis for scoring functions in predictive maintenance.

3.3.1 Process Monitoring System

The process monitoring system in this production environment collects and stores historical and live operational data from various monitoring components such as sensors, actuators, vision systems, and test stations. Sensors measure parameters such as temperature, pressure, part position, and alignment, while actuators include robot arms, grippers, AGVs, and conveyor drives. Vision systems capture inspection results and pass/fail rates, and test stations record electrical and dimensional checks.

Historical data was particularly valuable for identifying critical and recurring process deviations, which could indicate early signs of functional degradation. For example, a trend of increased variance in alignment readings might not yet trigger a machine stop, but could be linked to higher failure rates over time, which is an important leading indicator for predictive maintenance [12].

In addition to raw parameter data, the process monitoring system also records alarm data and machine stoppages. These records were used to cross-check whether certain failures recorded in the CMMS had corresponding abnormal trends or alarms in the process monitoring data. This cross-validation strengthened the reliability of the functional criticality analysis.

To preserve confidentiality, all examples in this thesis use mock parameter names such as “Sensor A” or “Actuator A” instead of actual component-specific tags.

3.3.2 CMMS (Maximo)

The CMMS data formed the second major input for this research. Maintenance work orders typically include a failure description, often written as free text by maintenance teams, identification of the faulty component (such as a sensor, actuator, or motor), the type of maintenance action performed (immediate corrective, deferred corrective, or preventive), the downtime duration, and the date and time of the event.

Because work order descriptions vary in detail and require a clear understanding, a manual review process was needed. Each work order was examined to identify the relevant component, determine which overview- and machine-level function it supported using the FAST breakdown, and then map the work order to the corresponding function for higher-level analysis. This manual mapping step was critical, as inconsistencies, spelling mistakes, and abbreviations in the raw data could otherwise lead to misclassification of functions or inaccurate scoring in the risk assessment.

3.3.3 Linking the Data Sources

The integration of process monitoring data and CMMS work order data was performed manually in this thesis, following a structured procedure. First, a failure event described in a CMMS work order, such as “Sensor A misalignment fault,” was identified. Next, the

corresponding function in the FAST diagram was located, based on the failed component. Historical process data was then reviewed to detect patterns leading up to the failure, such as increasing cycle times, repeated alarms, or gradual parameter drift. Finally, the relevant details, including failure frequency, downtime, and failure visibility, were recorded for use in the condition-based risk ranking.

Although this process was time-intensive, it ensured accurate mapping between physical process behaviour and functional performance. Importantly, it also served as the groundwork for the proposed machine learning framework, where the same linking process could be automated.

3.3.4 Advantages of Combining Process Monitoring and CMMS Data

The combination of these two data sources offered several advantages for predictive maintenance strategies. Cross-verification of failure events reduced the likelihood of false scoring during the criticality assessment. Process monitoring trends provided visibility into hidden degradation that might not yet appear in work orders, revealing early signs of deterioration. Linking process monitoring data to maintenance work orders also strengthened root cause reasoning by demonstrating why specific interventions were necessary. Finally, the integration of time-series process data with labelled failure events created a solid foundation for predictive analytics, enabling the application of machine learning in future studies [13].

3.4 Condition-Based Risk Ranking (Visibility, Frequency, Downtime)

Once each function from the FAST breakdown was linked to its corresponding maintenance records and process monitoring data, the next step was to determine how critical each function is to the battery module production line's overall performance. In this research, criticality was quantified using a condition-based risk ranking method built on three parameters:

1. Failure Visibility — how easily a failure can be detected during operation.
2. Frequency — how often the failure occurs within the recorded maintenance history.
3. Downtime — how much time is lost when the function fails.

This approach moves beyond purely theoretical asset criticality methods by integrating the assessment into actual operational history. It focuses on functional importance, rather than just component cost or replacement difficulty, which makes it better suited to predictive maintenance planning in complex automated systems.

3.4.1 Risk Parameters

3.4.1.1 Failure Visibility

Functions with low visibility (e.g., hidden sensor failure) may remain unnoticed for longer, causing downstream inefficiencies before being detected. Failures with high visibility (e.g., conveyor stoppage) are observed immediately, but may still be critical due to their impact on throughput.

3.4.1.2 Frequency

The historical recurrence of a failure is one of the strongest predictors of future behaviour. By analysing CMMS work order data over a particular time frame, functions with consistently

high failure frequency can be flagged as high-risk functions for predictive maintenance implementation.

3.4.1.3 Downtime

Downtime directly translates to production loss. Even low-frequency failures can be highly critical if each occurrence results in long recovery times. Downtime figures from CMMS records were used in combination with process monitoring logs to validate the impact of each failure [4].

These three parameters were chosen because they are directly observable from CMMS and process monitoring data, they can be audited, and they can be easily explained to engineering, maintenance, and operations teams without ambiguity.

3.4.2 Scoring Methodology

Each function identified in the FAST diagrams was evaluated against the three parameters using a standardised numerical scale. Failure visibility was rated from 1, representing failures that are immediately obvious, to 5, representing failures that are difficult to detect without detailed inspection or analysis. Frequency was scored based on the number of times the failure appeared in the work order history during production, ranging from 1 for rare occurrences to 5 for very frequent failures. Downtime was assigned a score based on the average production loss per failure, with 1 representing minimal downtime and 5 representing extended stoppages.

The overall risk score for each function was calculated as:

Overall Risk Score = Failure Visibility * Frequency * Downtime

The overall risk score for each function was calculated by multiplying the three individual parameter scores. This produced a raw score between 1 and 125, which was then standardised to a value between 1 and 5 for easier comparison.

Tabel 1: Standardised Risk Score

Overall Risk Score	Standardised Risk Score
1-25	1
26-50	2
51-75	3
76-100	4
101-125	5

Functions with a standardised risk score of 4 or 5 were classified as high priority for further review, sensor adequacy analysis, and potential integration into predictive monitoring strategies.

Example

The following table illustrates the scoring method used on the overview level:

Tabel 2: Scoring Method used on the overview level

Work order	Function (Overview Level)	Visibility	Frequency	Downtime	Overall Risk Score	Standardised Risk Score

No sensor signals detected	Acknowledge part presence (Machine A)	5	5	4	100	4
Gripper stopped	Move robot (Machine B)	3	5	3	45	2
Cell damage	Actuate cell gripper (Machine C)	2	3	5	30	2

In the table above, the "Acknowledge part presence" function in machine A has the highest standardised risk score because it is both frequent and failure visibility is very low, because detection happens after the component fails, while "Move robot" in machine B has a better failure visibility but fails more often, leading to a standardised risk score of 2. "Actuate cell gripper" in machine C has a good failure visibility but causes long production stoppages, making it critical despite its lower frequency.

3.4.3 Stage 1 — Overview-Level Mapping

The first stage of the workflow focused on an overview-level mapping of maintenance data, primarily through the analysis of work orders in the CMMS. To begin, work orders from Maximo were exported and cleaned, with particular attention given to Immediate Corrective Maintenance (ICM) entries, as these are critical. While ICM was prioritised, other types of maintenance, such as Deferred Corrective Maintenance (DCM), Preventive Maintenance (PM), and Emergency Maintenance (EM), were retained to provide additional context. As part of this preparation, problem descriptions were standardised to correct spelling or abbreviation errors, ensuring uniformity across the dataset.

Once the data was cleaned, each work order was manually mapped to the appropriate function in Excel. This required interpreting the free-text descriptions to identify the failed component, for example, "Sensor A misalignment" or "Replace Actuator B." The mapping ensured that every work order was aligned with the relevant overview-level function in Machine A, B, or C.

For each overview-level function, a structured record was then created in Excel, including details such as the work order reference, asset number, maintenance type, frequency score, failure visibility score, downtime score, and supporting notes from the process monitoring platform (for example, recurring alarms). The scoring, as mentioned in the defined parameter evaluation tables in section 3.4.2, was applied to assign frequency, visibility, and downtime scores to each function. To improve prioritisation, an overall risk score was also calculated by multiplying frequency, failure visibility, and downtime scores. Functions that received a standardised risk score of 4 or 5 were flagged as high-risk.

The outcome of this stage was a shortlist of high-risk overview-level functions. Only those functions meeting the threshold (standardised risk score ≥ 4) were retained, and this refined list provided the input for the second stage of analysis at the machine-level.

3.4.4 Stage 2 — Machine-Level Focus

The second stage focuses on the shortlisted high-risk functions by moving from the overview to the machine level. Each high-risk overview-level function was broken down into its corresponding machine-level sub-functions. For example, an overview-level function such as "Verify placement" could be broken down into machine-level components, including "Acknowledge component position," "Actuate Sensor A," and "Acknowledge gripper pressure." This allowed a more detailed view of where failures occurred within the operational hierarchy.

Work orders associated with the shortlisted functions were then re-examined to ensure that they were accurately mapped at the machine level. The descriptions of corrective actions and component failures were carefully cross-checked to verify that the correct function was identified. Parameter scoring, covering frequency, failure visibility, and downtime scores, was applied again at this more detailed level, mirroring the approach used in Stage 1.

Following scoring, the machine-level functions were ranked according to their criticality. Functions that reached a standardised risk score of 4 or 5 were classified as final critical functions, forming the basis for visualisation and reporting of the results. Where possible, this mapping and scoring were further validated by cross-referencing process monitoring data such as alarm trends, sensor performance, and the status of other monitoring signals. This cross-verification helped strengthen the reliability of the identified critical functions and ensured consistency between work-order analysis and actual process behaviour.

3.4.5 Scoring rules

Frequency score

Tabel 3: Frequency score based on the number of similar recorded occurrences

Number of occurrences	Score
1–2	1
3–4	2
5–6	3
7–8	4
>8	5

Frequency scores were assigned based on the number of similar recorded occurrences, ranging from 1 for 1 to 2 failures, to 5 for more than 8 failures within the assessment or specific period.

Failure visibility score

Table 4: Failure visibility score based on maintenance type

Maintenance type	Practical meaning (visibility context)	Score
Emergency Maintenance (EM)	Failure revealed at the point of crisis, safety/urgent impact; no advance visibility	1
Preventive Maintenance (PM)	Scheduled checks catch issues before failure; generally high visibility	2
Deferred Corrective Maintenance (DCM)	Failure is visible, but action is delayed; moderate visibility	3

Immediate Corrective Maintenance (ICM)	Failure becomes visible only upon occurrence; rapid response is needed; post-failure detectability	4

ICM is prioritised for the final risk ranking because these events directly affect the production rate and require rapid diagnosis to avoid production loss and cost escalations.

Downtime score

Tabel 5: Downtime score based on workorder

Downtime per work order	Score
< 30 min	2
0.5–1 h	3
1–1.5 h	4
> 1.5 h	5

DCM/PM maintenance types are assigned 1 for downtime, but not used in the final high-criticality assessment, as the focus remains only on ICM.

3.4.6 Critical Function Visualization

To make the results of the condition-based risk ranking accessible to both technical and managerial people, each function can be represented visually with a simple, colour-coded function that corresponds directly to its standardized risk score. The visualisation approach follows a standardized four-colour scale.

Functions with a score of 1 were shown in green, representing low risk and minimal maintenance needs. A score of 2 was shown in yellow, indicating moderate risk that requires occasional monitoring. Scores of 3 were shown in grey, representing higher risk functions that require active monitoring to avoid operational disruptions. Finally, scores of 4 and 5 were highlighted in red, indicating them as very high risk and requiring immediate attention to prevent major production losses.

This colour scale allowed quick prioritisation during review meetings or maintenance planning discussions, even without interpreting raw numbers. Additionally, a risk grid can be used to plot functions by Downtime (x-axis) and Frequency (y-axis), with bubble size representing Failure Visibility. In all cases, the colour coding remains consistent, ensuring that high-risk functions stand out among the other low-risk functions in both the overview and machine-level. This standardisation supports quick decision-making and reduces ambiguity when interpreting results.

3.4.7 Focus on Immediate Corrective Maintenance

The risk ranking in this thesis focused on ICM work orders because they directly influence production rate and operational efficiency. These events require rapid diagnosis and resolution to prevent production loss and cost escalation. They also provide the clearest basis for targeted monitoring improvements, as the cause-and-effect relationship between failure and resolution is typically well defined. While DCM and PM data are valuable for long-term reliability improvements, they were excluded from the final criticality scoring to maintain focus on immediate production risks.

3.4.8 Advantages

The main advantage of this approach is that it provides an objective and data-driven framework. Functions at both the overview and machine levels are ranked using quantifiable parameters, which gives concrete support for maintenance analysis. The numeric scoring system can be integrated into dashboards such as Power BI, CMMS plugins, or ML-based platforms in future implementations [14]. Condition-based Risk Ranking and Risk-based Prioritization are common in PdM.

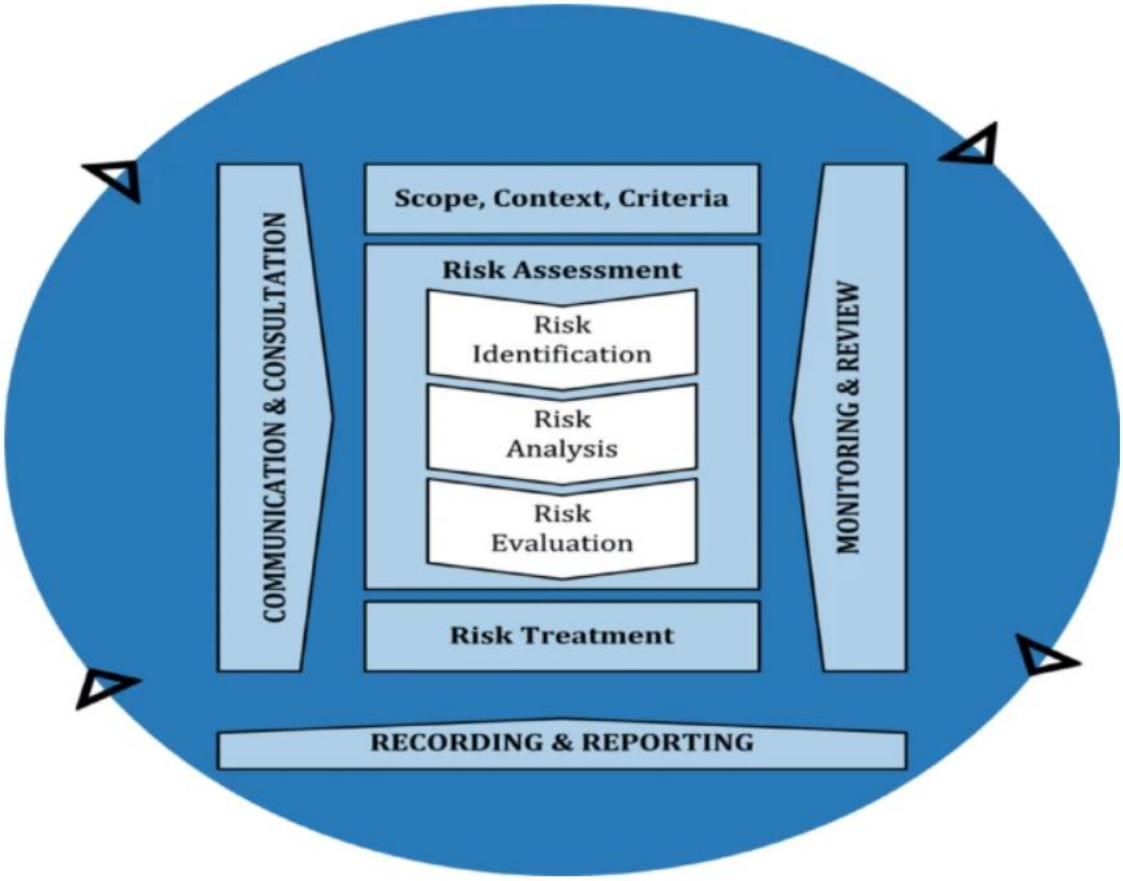


Figure 2: Multi-Criteria Risk Assessment [15]

The standardised risk score parameters can also be adjusted for different maintenance environments. For example, downtime may be weighted higher for just-in-time operations. This flexibility enables maintenance teams to prioritise sensor upgrades, develop customised strategies, and stock spare parts for the highest-risk functions with greater clarity.

3.4.9 Limitations

However, the condition-based risk ranking approach is not without limitations. Manual mapping of free-text work orders to relevant overview- and machine-level functions is time-consuming and subject to interpretation errors, particularly when descriptions are incomplete. To mitigate this, a standardised vocabulary should be adopted, avoiding inconsistent grammar and spelling.

Another limitation is the exclusion of sensor-level FAST mapping, which means that some fine-grained diagnostic opportunities were missed. If introduced in future work, this would

enable traceable function IDs and tighter linkage between real-time PLC/SCADA signals and Maximo, which is ideal for ML integration.

Nevertheless, the three-parameter scores, combined with time-series data from process monitoring (trends, alarm logs), provide a clean foundation for developing recommendation models. Such models could suggest which function is most likely to cause the next ICM and what intervention typically resolves it.

3.5 Sensor Coverage and Monitoring Capability

A reliable predictive maintenance framework depends not only on accurate fault classification but also on the quality and completeness of data obtained from sensors and monitoring systems. Even a well-defined functional breakdown and risk ranking process will underperform if the supportive sensory data does not adequately capture the operational state of critical functions.

This section evaluates the coverage and adequacy of sensors within the battery module production line, focusing on the functions identified as high-risk in Sections 3.3 and 3.4. The evaluation considers both the type and placement of sensors, as well as their ability to detect early signs of degradation.

3.5.1 Role of Sensor Coverage in Predictive Maintenance

Sensor coverage determines the observability of critical machine functions. In industrial automation, each function relies on a combination of measurement devices, such as contact and non-contact sensors, vision cameras, thermocouples, and pressure transducers, to report its operational state. The adequacy of these devices directly influences the ability to detect faults in time to prevent costly unplanned downtime.

In predictive maintenance, incomplete sensor coverage can miss early degradation trends, leading to late fault detection and reactive maintenance interventions. For example, a robotic gripper may have a proximity sensor confirming battery pickup, but without pressure monitoring, the system cannot detect a gradual grip-force decline. Conversely, redundant sensors, where two or more devices monitor the same parameter, may improve reliability but could also introduce unnecessary costs if not justified by risk analysis.

3.5.2 Method for Assessing Sensor Adequacy

The assessment in this study began by mapping high-risk functions, identified in the condition-based risk ranking, against the existing monitoring systems' availability. For each function, the corresponding machine-level sub-functions were reviewed to determine which sensors directly contributed to fault detection. This mapping process was carried out using the FAST diagrams and anonymised machine operations to maintain confidentiality.

The evaluation criteria were designed to assess whether the available sensors could:

1. Detect faults before they cause production stoppage.
2. Provide measurable indicators that can be trended over time for predictive modelling.
3. Operate reliably under environmental conditions such as vibration, temperature variation, and smoke.

For example, “Acknowledge part presence” in Machine A, the available inductive proximity sensor detected whether a component was in position. However, cross-referencing this with CMMS work order history showed recurring instances of robot stoppage, suggesting potential misalignment or insufficient sensing range in the robot arm. These issues indicated a gap in detection capability, particularly for early-stage misalignment faults that immediately cause stoppage and lead to frequent ICM.

3.5.3 Sensor Placement and Signal Quality

The physical placement of a sensor significantly influences its effectiveness. Misalignment, mounting at an improper distance, or obstruction by machine components can result in unreliable readings. Signal quality also depends on the integrity of cabling, shielding from electrical noise, and correct calibration. Inadequate signal conditioning, such as low sampling rates or insufficient filtering, can further reduce fault detection sensitivity.

In this research, assessment of signal quality was carried out indirectly by analysing patterns in process monitoring data and schematics in electrical documents. Stable sensor signals during known operational changes suggested reliable measurement, while excessive noise or sudden signal dropouts were flagged as possible indicators of hardware or wiring issues. These findings were linked back to the machine-level functions in the FAST breakdown to identify where corrective action could improve fault detection.

3.5.4 Identified Monitoring Gaps

The analysis revealed two main types of monitoring gaps. First, in certain high-risk functions, only binary (on/off) sensors were present, limiting the ability to track gradual parameter drift. Second, some critical functions lacked direct measurement altogether, relying instead on downstream indicators such as product rejection rates or operator intervention reports. While these additional indicators are useful, they introduce delays in fault detection and complicate root cause analysis.

Addressing such gaps requires strategic sensor upgrades. In the context of predictive maintenance, the introduction of IoT sensors, vibration monitoring devices, or AI-driven vision systems could enable richer datasets and earlier fault detection. However, such upgrades should be prioritised based on the condition-based risk ranking results from Section 3.4 to ensure cost-effective investment.

3.5.5 Implications for Predictive Maintenance Framework

Adequate sensor coverage serves as the foundation for implementing machine learning-based predictive maintenance. Without robust and comprehensive sensing, predictive models may suffer from incomplete feature sets, resulting in lower accuracy and false predictions. Conversely, targeted improvements to sensor coverage, supported by the results of the condition-based risk ranking, can significantly increase both the detection and prediction of failures.

In this thesis, while the evaluation did not extend to the sensor-level FAST mapping due to scope, the principles applied here could be extended in future work. A fully sensor-level breakdown would enable the creation of unique function identifiers linked directly to real-time PLC or SCADA signals, which would greatly contribute to automated fault mapping and predictive model training [14].

3.6 Integration of FAST, Risk Ranking, and Sensor Coverage

A predictive or smart maintenance framework for a highly automated production system, such as Scania's battery module line, cannot rely on a single method of assessment. Instead, the methodology developed in this thesis integrates three complementary layers: (1) functional mapping through FAST, (2) quantitative prioritisation via condition-based risk ranking, and (3) evaluation of monitoring capability through sensor coverage assessment. Together, these layers form a structured approach that transforms raw operational data and qualitative maintenance records into actionable insights for predictive maintenance planning.

3.6.1 Linking Functional Breakdown with Risk Ranking

The FAST methodology provided the structural backbone for understanding how the production line operates. By mapping both overview-level and machine-level functions across Machines A, B, and C, it became possible to connect maintenance records to specific functions rather than treating failures only at a component or machine level. Once this mapping was established, the condition-based risk ranking method was applied to quantify the relative criticality of each function using failure visibility, frequency, and downtime.

For example, in Machine A, the function *"Acknowledge part presence"* was linked to multiple ICM work orders where proximity sensors in the robot failed to detect incoming cells. This was identified as a critical function because of its high recurrence rate and low visibility. By contrast, the function *"Actuate gripper"* in Machine C, although less frequent, was scored highly because failures resulted in long stoppages and significant production loss. These findings demonstrated that different functions can be critical for different reasons, highlighting the need for a combined structural and data-driven approach.

3.6.2 Incorporating Sensor Adequacy into the Risk Framework

The risk ranking alone identifies which functions are most critical, but it does not explain whether the monitoring infrastructure is capable of detecting early degradation. The sensor coverage assessment addressed this by overlaying monitoring availability on top of the risk results.

In practice, this meant examining whether high-risk functions had sufficient sensors to capture degradation trends. For instance, while Machine B's robotic pick-and-place function was equipped with proximity sensors, the CMMS history suggested recurring misalignment in the cell. Review of the process monitoring data revealed that these sensors only provided binary confirmation signals, meaning that gradual drift leading to misalignment was not visible. This showed that the function was both high-risk and need additional support, creating a strong justification for targeted sensor upgrades.

3.6.3 Prioritisation Workflow for Predictive Maintenance

By integrating the three methodologies, a prioritisation workflow was developed for Scania's battery module line. The process began with a FAST breakdown, where both overview- and machine-level functions were defined. Once this structural mapping was established, risk scoring was carried out by evaluating each function against failure visibility, frequency, and downtime using CMMS and process data.

Functions with scores of four or higher were then shortlisted as high-risk and selected for deeper review. For these critical functions, an additional layer of analysis was applied to assess sensor adequacy, determining whether existing monitoring systems provided sufficient coverage. Based on the combined insights from functional analysis, risk scoring, and sensor evaluation, targeted interventions were recommended. These included predictive monitoring measures, sensor upgrades, or adjusted maintenance strategies.

This workflow ensures that maintenance resources are directed not only to the most critical functions but also to those with the weakest monitoring capability. In this way, predictive maintenance becomes both effective and reliable.

3.6.4 Challenges of Integration

The integration process was not without challenges. Manual mapping of free-text work orders to functions required careful effort and was occasionally limited by incomplete descriptions. Furthermore, while the FAST model provided logical clarity, it did not capture sensor-level detail within the scope of this thesis. This limited the ability to automate mapping between real-time sensor signals and function IDs. Nevertheless, the manual approach demonstrated feasibility, interoperability capabilities in their existing machines and control systems, and created a small workflow that could serve as learning material for future machine learning models.

Another challenge was balancing confidentiality with analytical depth. Machine operations and functions had to be anonymised, which limited the specificity of certain examples. However, this did not undermine the methodology itself, as the process is transferable and can be replicated across other automated production environments.

3.6.5 Implications for Scania's Battery Module Line

The integrated approach developed in this thesis has clear implications for Scania's battery production. By linking functional structure, operational risk, and monitoring adequacy, it becomes possible to pinpoint functions that drive most ICM work orders and production downtime. Identifying where sensor gaps prevent early fault detection and providing a prioritised roadmap for predictive maintenance implementation.

For a system as complex as a battery module line, where robots, conveyors, vision systems, and testers interact in tightly coupled sequences, this integration ensures that predictive maintenance targets the areas of greatest vulnerability. In the long term, extending this approach to a sensor-level FAST breakdown and integrating it with MES, SCADA, process monitoring, and CMMS platforms could enable automated fault classification and near-real-time predictive alerts.

4. Results and Analysis

4.1 Functional Risk Scores

The first stage of analysis focused on quantifying the relative criticality of functions across the battery module production line. Using the framework described in Section 3.4, each function identified in the FAST diagrams was linked to the corresponding ICM work orders in Maximo. The three parameters, failure visibility, frequency, and downtime scores, were applied to the

dataset, and the resulting scores were used to identify functions most critical to production flow.

At the overview level, each major process function, such as component verification, robotic actuation, or gripper movement, was assigned a set of scores based on work order history. These values were then standardised on a scale from one to five, allowing comparison across functions with very different operational roles. Once these overview functions were assessed, those that scored four or five were filtered forward to the next stage of machine-level analysis. This two-step evaluation made it possible to first isolate broad areas of concern and then focus on the specific machine-level functions most responsible for stoppages and recurring failures.

Table 6 presents an illustration of the results obtained during the overview-level analysis. Three representative examples are included to demonstrate how the condition-based risk ranking method differentiated functions.

Tabel 6: Overview-level functional risk scores

Work Order Description	Function (Overview Level)	Visibility	Frequency	Downtime	Overall Risk Score	Standardised Risk Score
Sensor A signal is not detected	Acknowledge part presence (Machine A)	5	5	4	100	4
Gripper stopped	Move robot (Machine B)	3	5	3	45	2
Cell damage	Actuate cell gripper (Machine C)	2	3	5	30	2

From this analysis, it was evident that the "*Acknowledge part presence"* function in Machine A presented the highest criticality. This function not only experienced frequent failures but also had low failure visibility, as detection occurred only after the fault had already caused a stoppage. With a standardised score of four, it was categorised as a high-risk function.

By contrast, the "*Move robot"* function in Machine B had a high failure frequency but was easier to detect and resolve, giving it a lower overall ranking. The "*Actuate cell gripper"* function in Machine C represented a relatively low frequency of failures but was associated with long downtime per incident, reflecting a different but equally relevant form of risk.

The next stage of the analysis moved from this overview to the machine level. Here, each shortlisted function was split into its constituent machine-level sub-functions, which were then reassessed using the same parameters. This enabled the study to determine not just which general areas of the battery module production line were most vulnerable, but which precise functions required targeted monitoring or additional sensor coverage.

4.2 Critical Function Mapping

While the overview-level results provided an understanding of which functions carried the greatest operational risk, predictive maintenance planning requires greater granularity. For this reason, the shortlisted high-risk functions identified in Section 4.1 were expanded into their

corresponding machine-level sub-functions. This step not only highlighted the root causes of repeated failures but also clarified which specific components were responsible for production losses. By linking machine-level functions to the FAST diagram and CMMS work orders, the risk assessment could move from general trends to actionable insights for Scania's battery module production line.

Table 7 illustrates this machine-level scoring for selected high-risk overview functions. For example, the function "Acknowledge part presence" in Machine A was broken down into sub-functions, including "Read inductive sensor," "Confirm robot arm alignment," and "Verify PLC signal handshake." Each of these was evaluated for failure visibility, frequency, and downtime using the same scoring methodology as in Section 3.4. Similarly, the function *"Move robot"* in Machine B was expanded into specific motion-related tasks such as "Actuate axis motor," "Engage gripper movement," and "Synchronise with conveyor."

Tabel 7: Machine-level risk scores for high-risk overview functions

Overview Function (Machine)	Machine-level Function	Visibility	Frequency	Downtime	Overall Risk Score	Standardised Risk Score
Acknowledge part presence (Machine A)	Read inductive sensor A	5	5	4	100	4
	Acknowledge robot arm alignment	4	4	3	45	3
	Verify PLC 1 signal handshake	3	3	4	30	3
Move robot (Machine B)	Actuate axis motor B	3	5	3	45	3
	Engage gripper movement	4	4	4	64	4
	Synchronise with conveyor 1	3	3	3	21	2

The machine-level results reveal important differences that were hidden at the overview stage. In the case of Machine A, the inductive sensor A, responsible for detecting component presence, consistently recorded a high failure rate and poor visibility, leading to a standardised score of 4. This confirms that the repeated stoppages linked to this function in the CMMS were primarily driven by sensor limitations, such as short detection range or misalignment, which had already been observed during the monitoring gap assessment in Section 3.5. On the other hand, the PLC handshake function, while not failing as often, still caused long recovery times when issues occurred, which increased its downtime score.

For Machine B, the robot's axis motor B showed the highest frequency of failures, but because the stoppages were generally resolved within 30–60 minutes, its standardised score remained moderate. In contrast, failures in the gripper's engagement mechanism scored higher because

they combined poor visibility with long recovery times, directly affecting throughput. The synchronisation step between robot C and conveyor 1 showed relatively balanced scores, but its lower frequency reduced its criticality compared to the other sub-functions.

Together, these findings show that different machine-level sub-functions within the same overview function can carry very different risk values. For predictive maintenance planning, this is crucial because it prevents broad interventions and instead directs attention to the exact functions that threaten continuity. At the Scania battery module line, the results suggest that sensor-based improvements in Machine A and mechanical adjustments in Machine B's gripper subsystem should be prioritised ahead of less critical functions and support tailored recommendations to avoid failures in the future. This machine-level mapping, therefore, transforms the abstract risk ranking into concrete maintenance priorities that can be linked directly to investment decisions, sensor upgrades, or early predictive model development.

4.3 Sensor Adequacy Results

The evaluation of sensor coverage at the Scania battery module line confirmed that several of the high-risk functions identified through condition-based risk ranking suffered from monitoring gaps. These gaps were not uniform across machines and stations but varied depending on whether the function relied on binary on/off signals, continuous real-time parameter monitoring, or indirect indicators. By systematically linking CMMS work orders with available process monitoring signals, it was possible to distinguish between failures caused by internal hardware issues and those worsened by insufficient sensing capability.

In Machine A, the overview-level function *"Read inductive sensor A"* showcased the clearest example of inadequate coverage. While the proximity sensor correctly detected part presence in most cases, its limited sensing range and sensitivity to misalignment caused recurring stoppages. This was reflected in both the CMMS records and the process monitoring data, where multiple instances of *"no part detected"* alarms were logged despite the part being physically present. The lack of supplementary measurements, such as pressure or distance sensing, meant that these preliminary errors could not be identified until the failure had already stopped production. This aligns with recent literature findings that highlight how binary sensors, although cost-efficient, are often insufficient for predictive maintenance in high-throughput assembly lines.

In Machine B, a different challenge emerged. Failures in the *"Engage gripper movement"* function were repeatedly linked with weak or unstable signals from the gripper's position sensor. Electrical schematics and process monitoring records indicated that electrical noise reduced signal reliability, resulting in downtime while technicians verified alignment and rewired connectors. In this case, the sensor type itself was adequate for detecting the motion, but issues with placement and signal quality compromised its effectiveness. Research in industrial maintenance confirms that insufficient sensor installation and environmental interference are among the most common root causes of false alarms and missed detections.

In Machine C, a broader monitoring gap was observed, where certain functions relied primarily on downstream indicators such as cell rejection rates or operator interventions. For instance, failures in the *"Actuate cell gripper"* function often appeared only after defective modules were flagged, by which time the failure had already stopped production. The absence of direct force sensing limited the ability to detect the failure trend in the gripper mechanism. While

downstream indicators provide some insight, they delay diagnosis and complicate root cause analysis, an issue that has also been reported in other automated battery assembly systems.

Taken together, these results demonstrate that the monitoring capability at Scania's battery module line is uneven across functions. Some critical functions are well-supported by sensors and vision systems but constrained by installation and calibration quality, while others rely only on binary or indirect indicators that limit predictive capabilities. Addressing these issues requires a targeted approach, where sensor upgrades are prioritised based on the condition-based risk ranking, rather than through untargeted investments. This ensures that resources are directed towards functions where improved sensing will deliver the greatest impact on production continuity.

By combining the machine-level risk assessment with the sensor adequacy evaluation, the study demonstrates how predictive maintenance planning can progress from identifying critical functions to addressing the underlying monitoring weaknesses. This creates a foundation for future predictive models that depend on robust and reliable data paths.

4.4 Discussion

4.4.1 Linking Risk Ranking to Sensor Adequacy

The combined use of condition-based risk ranking and sensor adequacy evaluation provides a clearer picture of where predictive maintenance efforts and strategies should be directed in the Scania battery module line. The results from Sections 4.1 and 4.2 highlighted critical functions with high-risk scores, particularly those linked to failures in part detection and robotic gripping. Section 4.3 then showed that many of these same functions also suffered from incomplete or unreliable sensor coverage.

This overlap is significant because it demonstrates that the most critical failures are not simply the result of mechanical wear but are aggravated by insufficient monitoring capabilities. Literature on predictive maintenance stresses that sensor reliability and placement are as important as the algorithms interpreting their signals, since data gaps directly affect model accuracy and timeliness [16].

4.4.2 Implications for Production Continuity

For a high-throughput line such as Scania's battery module assembly, downtime events spread quickly across upstream and downstream processes. The identification of Machine A's proximity sensor gap, for instance, shows that even small detection failures can escalate into major stoppages. Similarly, the signal instability observed in Machine B's gripper subsystem highlights the impact that certain electrical installation practices can have on operational reliability, suggesting an opportunity for improvement to prevent future stoppages.

When such failures are mapped onto the risk grid, their impact on overall line continuity becomes evident. Studies in automated assembly systems have reported that undetected or late-detected failures can reduce overall equipment effectiveness (OEE) by more than 10% in comparable environments. This finding highlights the importance of linking functional risk ranking with real-world monitoring data rather than treating these aspects separately.

4.4.3 Value of Targeted Interventions

One of the most important outcomes of this research is the demonstration that predictive maintenance cannot be implemented effectively through generic sensor upgrades. Instead, investments must be targeted to the most critical and least observable functions.

In the case of Scania's module line, upgrading the sensing capability of Machine A's part detection or introducing improved noise filtering for Machine B's gripper sensors would provide greater returns than broad upgrades across all stations. This is consistent with recent industry findings that motivate risk-driven sensor deployment to avoid cost escalation and maximise reliability.

Furthermore, by focusing on functions flagged by the condition-based risk model, engineering teams can integrate predictive monitoring into existing workflows with fewer disruptions and clearer justifications.

4.4.4 Foundation for Predictive Modelling

Another key implication is that high-quality sensing forms the foundation for predictive analytics and machine learning. If critical functions are monitored only by binary sensors or downstream quality indicators, the ability to detect failure trends before failure is greatly reduced. This restricts the usefulness of predictive models and increases reliance on reactive responses.

The results of this study suggest that Scania's current monitoring setup, while effective for immediate detection, requires expansion into richer data streams such as force, vibration, noise, and pressure signals for the most critical functions. Integrating these improvements would create a more robust dataset that can be used in advanced AI-driven maintenance frameworks, as demonstrated in recent predictive maintenance applications across the automotive sector [14] and [17].

4.4.5 Broader Lessons for Industry 4.0 Maintenance

Finally, the findings extend beyond the Scania battery plant by reinforcing the principle that predictive maintenance is not achieved through algorithmic solutions alone. The foundation lies in understanding functional criticality, evaluating monitoring gaps, and then systematically closing those gaps with targeted sensor strategies.

This sequential approach reflects the broader Industry 4.0 shift from traditional maintenance scheduling to data-driven interventions. It also aligns with current trends in smart manufacturing, where cross-linking CMMS records, FAST diagrams, and real-time sensor data enables a more holistic view of asset health [18] [4].

4.5 Limitations

Although this research provides a structured approach to linking condition-based risk ranking with monitoring system adequacy, several limitations must be recognised.

A key limitation was the restricted availability of data. While CMMS work orders and process monitoring records formed the backbone of the analysis, access to certain qualitative and quantitative maintenance information was limited. Specifically, detailed quality deviation logs, historic camera data, and complete sensor histories were either unavailable or still under

integration at the time of the study. This reduced the possibility of capturing long-term degradation trends and limited the ability to fully validate the condition-based ranking results. Similar challenges are common in industrial predictive maintenance research, where partial datasets often limit analysis and mandate the use of approximations [17].

Another limitation lies in the reliance on historical CMMS work orders as the primary reference for failure visibility, frequency, and downtime. While these records are valuable, they are subject to the consistency of technician reporting and the quality of free-text entries. Variations in how failures are described, including inconsistent terminology or abbreviations, required manual cleaning and interpretation. This manual intervention introduces the potential for human error, a known challenge when using CMMS data in predictive maintenance studies [4].

The condition-based risk ranking itself was defined using three parameters: failure visibility, frequency, and downtime. These were chosen for their practical observability and ease of explanation to operations teams, but other important factors, such as safety, energy use, and spare part availability, were not included. While the narrowed scope ensures clarity and comparability, it also means the analysis does not capture all possible dimensions of functional criticality. Extending the scoring system with multi-criteria decision-making approaches could address this in future work.

A further limitation is that sensor adequacy assessment was carried out at the overview and machine-function level, rather than at the sensor-signal level. This meant issues such as calibration changes, sampling frequency, or electrical interference were not systematically evaluated. Previous research shows that such low-level data quality factors can significantly influence predictive model accuracy.

Finally, this thesis did not include the implementation or validation of a machine learning model. While the methodological framework is designed to support future predictive modelling, its effectiveness has not yet been tested in a live production environment. This was limited to a scope decision, with the focus on establishing a structured foundation for future ML-based work.

4.6 Conclusion

This research aimed to demonstrate how a function-based approach can be used to prioritise maintenance planning on Scania's battery module lines in a way that is both defensible and immediately useful. Starting from a FAST breakdown, the analysis linked overview functions to machine-level sub-functions and then compared those findings against operational history retrieved from CMMS work orders and process monitoring signals.

The condition-based risk ranking transformed that history into comparable scores for failure visibility, frequency, and downtime, allowing high-risk functions to be filtered without ambiguity. The subsequent deeper breakdown at the machine level clarified which specific tasks, rather than whole machines, were responsible for stoppages. Finally, the sensor or monitoring system adequacy assessment explained why some failures persisted: not only because of mechanical or control issues, but also because key functions lacked the

measurements needed to expose failure before it became an Immediate Corrective Maintenance activity.

The results highlighted a consistent pattern across the line. Functions tied to part presence and robotic gripping accounted for a disproportionate share of production interruptions. In Machine A, repeated *"no part detected"* stoppages aligned with a limited sensing range and sensitivity to misalignment. In Machine B, gripper movement issues were traced back to signal reliability rather than the choice of sensor type. Machine C revealed a different weakness: several functions were visible only through downstream quality effects, which delayed diagnosis and complicated root-cause analysis.

These findings are significant operationally because they convert a broad "high risk" label into concrete actions: improve early-stage detection where the signal is binary, strengthen installation and noise immunity where signal stability is poor, and introduce direct condition measurements where only indirect indicators exist.

A practical implication for Scania is that predictive maintenance investment should be sequenced by function, not by asset class. The evidence supports targeted upgrades, additional real-time analogue sensing, such as IoT sensors for alignment and grip force where drift leads to stoppage, adequate but calculated improvements to wiring and shielding on motion feedback where electrical noise has been observed, and selective use of vision or displacement sensing where current checks are indirect.

Because these priorities come from traceable links, FAST mapping, risk scoring, and monitoring review, they can be implemented within existing governance, justified in maintenance planning meetings, and measured against future reactive maintenance, such as ICM and DCM, work order reductions.

Methodologically, the thesis also establishes a foundation for smarter decision support. The structured mapping between work orders and functions, the parameterised risk scores, and the notes recorded during manual mapping form a supportive path for future automation. With better monitoring on the most critical functions, the same workflow can be implemented into a data pipeline for predictive models, without changing the underlying logic that operations and maintenance teams have already validated.

The limitations noted earlier set the boundaries of what can be claimed. Partial access to quality deviations, historic camera data, and complete sensor histories constrained how far trend analysis could go, and reliance on free-text work orders required careful cleaning and interpretation. Even so, the integrated approach proved sufficient to identify where monitoring was strong, where it was weak, and where adequate, targeted changes would improve uptime.

Overall, the research shows that a function-based, data-driven methodology can prioritise actions that protect throughput on an automated battery module line. It replaces generic discussions about "more sensors" or "more AI" with a focused, defensible sequence: mapping functions, risk score from actual available history, and verifying whether the monitoring in place can genuinely see failure early enough to act. That sequence is scalable, compatible with

Scania's existing systems, and ready to be extended in the next phase toward predictive models and decision support that operate on the same functional backbone.

5 Future Scope

5.1 Machine Learning Introduction

The shift towards autonomous, highly automated production systems, such as those deployed in Scania's battery factory, demands maintenance strategies that are proactive, tailored, data-driven, and scalable. Traditional maintenance strategies, while effective in more manually supervised environments, are not sufficient for advanced production systems where robots, conveyors, sensors, and vision systems operate with minimal human intervention. In such contexts, failures that go undetected for even short periods can escalate into production stoppages, quality losses, and increased operating costs [14].

The functional analysis and condition-based risk ranking methods applied in this thesis explain a structured way to identify the critical functions and highlight where monitoring gaps exist. However, these methods required manual interpretation of maintenance records, cross-checking of CMMS work orders, and careful mapping against functions. This manual effort, while effective for a limited dataset, is not sustainable when scaled to hundreds or thousands of functions across multiple production lines. Moreover, free-text work orders in CMMS often contain inconsistencies, spelling errors, or short descriptions, which increases the complexity of systematic analysis [5].

Machine learning (ML) offers an opportunity to automate and extend this assessment. ML algorithms are capable of learning from historical data patterns to classify, predict, and recommend actions. In predictive maintenance, models can be trained to predict failures, estimate remaining useful life, or suggest corrective measures by combining structured time-series sensor data with unstructured maintenance logs. For Scania's battery module line, integrating ML into the framework would mean that functions identified as high-risk could be monitored continuously and linked with real-time alerts, while maintenance teams receive data-driven recommendations or suggestions on how to address specific failures.

A further consideration for ML adoption lies in the interoperability between CMMS, process monitoring platforms, and advanced analytics with Power BI. By leveraging AI techniques such as retrieval-augmented generation (RAG), maintenance systems could automatically retrieve similar past work orders, summarize their root causes, and suggest corrective recommendations based on proven interventions. This would address one of the most time-consuming aspects of maintenance decision-making, interpreting historical records and transferring inferred knowledge from experienced technicians to newer staff.

In summary, while functional breakdown and risk ranking provided the necessary foundation for understanding criticality, ML provides the means to implement and scale predictive maintenance in complex, robotic production environments. The next sections of this chapter outline how Maximo work orders can be used as a use case for ML training, explore potential model types for classification and recommendation, and propose a conceptual integration framework tailored for Scania's production setting.

5.2 Machine Learning Model Suggestion

Building the manual workflow described earlier, ML provides the tools to automate three essential tasks in the predictive maintenance framework, which interpret free-text CMMS work orders, predict functional criticality, and recommend corrective actions. Each of these tasks reflects the steps explored in this research, but the use of ML makes them scalable and less dependent on manual interpretation.

The first layer of this framework is focused on text classification of work orders. Maximo records typically contain short and sometimes inconsistent free-text work order descriptions, which describe the symptom of a fault or the component affected. Natural Language Processing (NLP) methods can be applied here to standardise and classify this unstructured text. Transformer-based models such as BERT, which have been successfully used in industrial data log analysis, can identify key terms, correct for variations in spelling or abbreviations, and map descriptions to predefined functions. For example, work orders describing “sensor fault,” “no signal,” or “sensor misaligned” can all be mapped under a single function label, increasing accuracy. Current research has reported that NLP can reduce manual interpretation time by up to one-third. while improving classification accuracy in maintenance datasets [19].

The second layer is “functional risk prediction”. Once work orders are standardised and mapped to functions, supervised ML models can be used to estimate the criticality of each function. This involves combining features from work orders (maintenance type, downtime, recurrence rate) with data from the process monitoring system (historical alarms, deviations, parameter changes). Algorithms such as Random Forests, Gradient Boosted Trees, or deep learning classifiers can be trained on these features to predict whether a function is likely to be high-risk in the future. In similar industrial studies, hybrid models that combined sensor data with CMMS logs improved prediction accuracy by 20–25% compared to using either dataset alone [9]. For Scania’s battery plant, this would mean that the model could flag, for example, that a specific gripper movement in Machine B is entering a high-risk state, based on recurring noise in the position sensor signal and past stoppages.

The third layer involves recommending corrective actions. Beyond classification, ML can suggest which interventions are most likely to resolve the problem, based on historical resolutions. For instance, if recurring failures of an inductive sensor in Machine A were consistently solved by recalibration or sensor replacement, the model can recommend the same corrective action when similar symptoms appear again. Such recommendation engines have proved to shorten technician diagnosis time and improve maintenance efficiency in complex automated environments [14].

Together, these three layers form a practical ML model suggestion: 1) use NLP to classify free-text work orders into structured function labels, 2) apply supervised learning to predict criticality scores based on both CMMS and sensor data, and 3) employ a recommendation system to suggest corrective actions based on past resolutions. Significantly, these models would not replace human expertise but act as a decision-support tool, allowing maintenance engineers to focus on validation and higher-level problem solving.

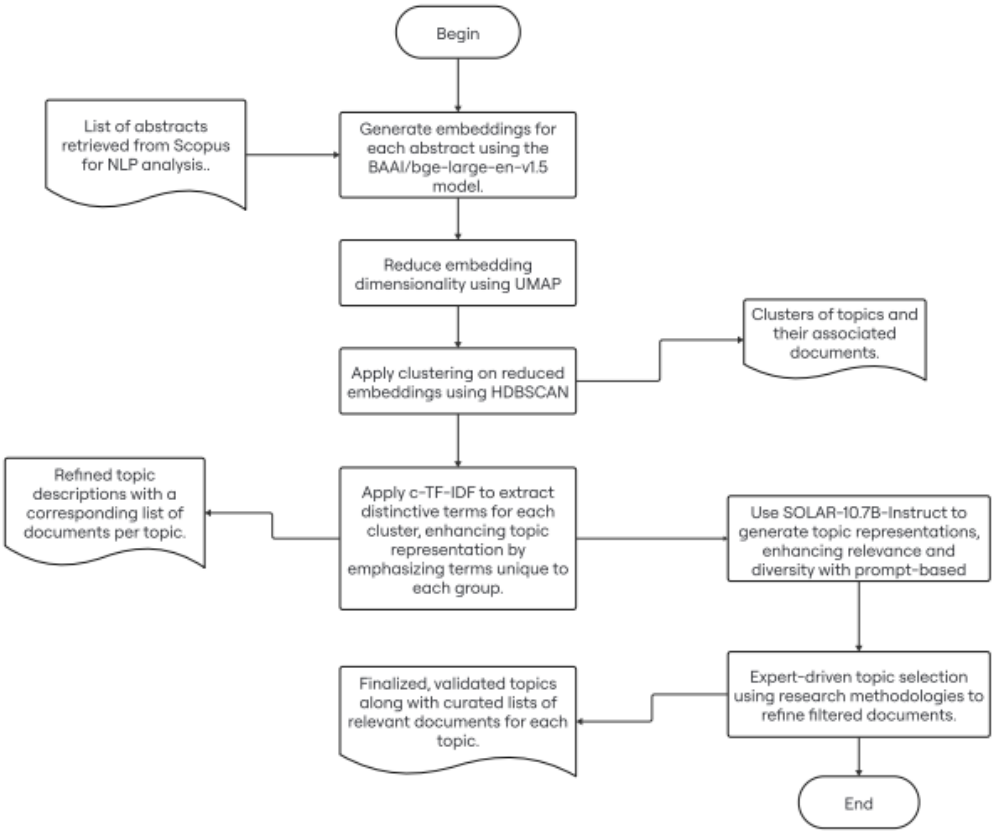


Figure 3: Flowchart representing an NLP-Based Framework [20]

By automating what has so far been performed manually, ML ensures that predictive maintenance becomes scalable across Scania's production lines. It also creates the foundation for interoperability with future AI-driven approaches, such as retrieval-augmented generation (RAG), where models could retrieve and summarise similar past cases automatically. This evolution is essential in complex, robotic, and highly automated production systems, where even minor stoppages can escalate into significant production and cost losses.

5.3 Integration for Predictive Decision Support

Integrating the proposed machine learning framework into Scania's existing maintenance ecosystem requires a practical and realistic design that aligns with the current workflow. At present, maintenance data are stored and managed through the CMMS, while process monitoring data are captured continuously by the line's digital monitoring platform. These two systems operate largely in parallel, and much of the analysis is conducted manually by engineers who review work orders, cross-check alarms, and interpret patterns. The goal of integration is to make these systems interoperable and capable of sharing information automatically, forming the basis of a decision-support loop for predictive maintenance.

In the proposed architecture, the data layer serves as the foundation. It consists of data pipelines that connect the CMMS, the process monitoring platform, and historical sensor logs. Each incoming work order from Maximo, whether preventive or corrective, is parsed and stored in a structured database together with its downtime, maintenance category, and timestamp. Simultaneously, process data such as alarm events, parameter deviations, and trend logs are

streamed or periodically exported into the same environment. This combined dataset represents the “single source of truth” from which machine learning models can learn and make predictions.

Above this, the analytics layer hosts the ML modules described previously. The text classification model processes free-text work orders and converts them into structured function labels, while the predictive model estimates the probability of failure for critical functions identified in the FAST analysis. When the probability of failure for a function exceeds a defined threshold, the model automatically triggers a recommendation request. The recommendation engine retrieves similar historical work orders, extracts their resolutions, and presents suggested corrective actions. This retrieval process can be enhanced using RAG methods, which allow the system to dynamically search and summarise relevant past events.

The visualisation layer is where the system interfaces with the maintenance team. The results of the ML predictions are displayed in a dashboard, which may be implemented through Power BI or directly integrated into the Maximo interface. Each function from the FAST diagram appears as a colour-coded tile reflecting its predicted risk level: green for stable, yellow for moderate risk, and red for high or critical. Selecting a tile provides a summary of recent alarms, correlated process parameters, and the recommended corrective actions. This interface translates complex analytics into information that technicians and engineers can interpret quickly, supporting faster and more informed decision-making.

A key advantage of this integration is the establishment of interoperability between systems that previously functioned in isolation. By linking Maximo and the process monitoring platform, data consistency improves, and manual duplication of analysis is reduced. Over time, as more work orders and process signals are collected, the model continues to learn and refine its predictions, creating a feedback loop between maintenance outcomes and predictive accuracy. Such an integrated setup not only strengthens the predictive maintenance capability at Scania’s battery factory but also supports long-term digital transformation towards autonomous maintenance planning.

To ensure scalability, this framework can later be deployed using edge-computing solutions that run directly on local servers within the factory network. This design keeps sensitive production data within Scania’s infrastructure, addressing data privacy concerns while maintaining real-time responsiveness. Conceptually, this mirrors modern industrial AI systems, where local analytics engines process data close to the source while synchronising key insights to central dashboards [21].

5.4 Limitations and Challenges of the Machine Learning Framework

The machine learning framework proposed in this thesis represents a realistic first step toward automating maintenance assessment at the Scania battery module line. Its primary value lies in demonstrating how existing data sources—CMMS work orders, process monitoring trends, and functional breakdowns—can be connected in a way that supports predictive decision-making. While the underlying architecture is feasible with Scania’s current digital infrastructure, several considerations shape its short-term applicability.

From a technical perspective, the main limitation relates to the availability and completeness of historical data. Although Maximo provides detailed maintenance logs, inconsistencies in free-text reporting and the absence of fully consolidated sensor histories restrict the depth of model training. ML models, particularly classification and recommendation systems, rely on large and diverse datasets to generalise effectively. In the current state, the framework would perform well for frequent or well-documented failure modes but may struggle with rare or newly emerging faults. As more integrated data becomes available, especially time-synchronised sensor trends and alarm logs, the reliability of predictions is expected to improve substantially.

Another challenge concerns the complexity of mapping functions to sensor signals. The ML design assumes access to well-labelled data linking functions, failures, and resolutions. In practice, this mapping still requires human interpretation to bridge the gap between engineering context (FAST functions), process behaviour, and textual descriptions in CMMS. Until a sensor-level breakdown or automated tagging system is introduced, the learning process remains partly dependent on manual preprocessing, which can slow down implementation.

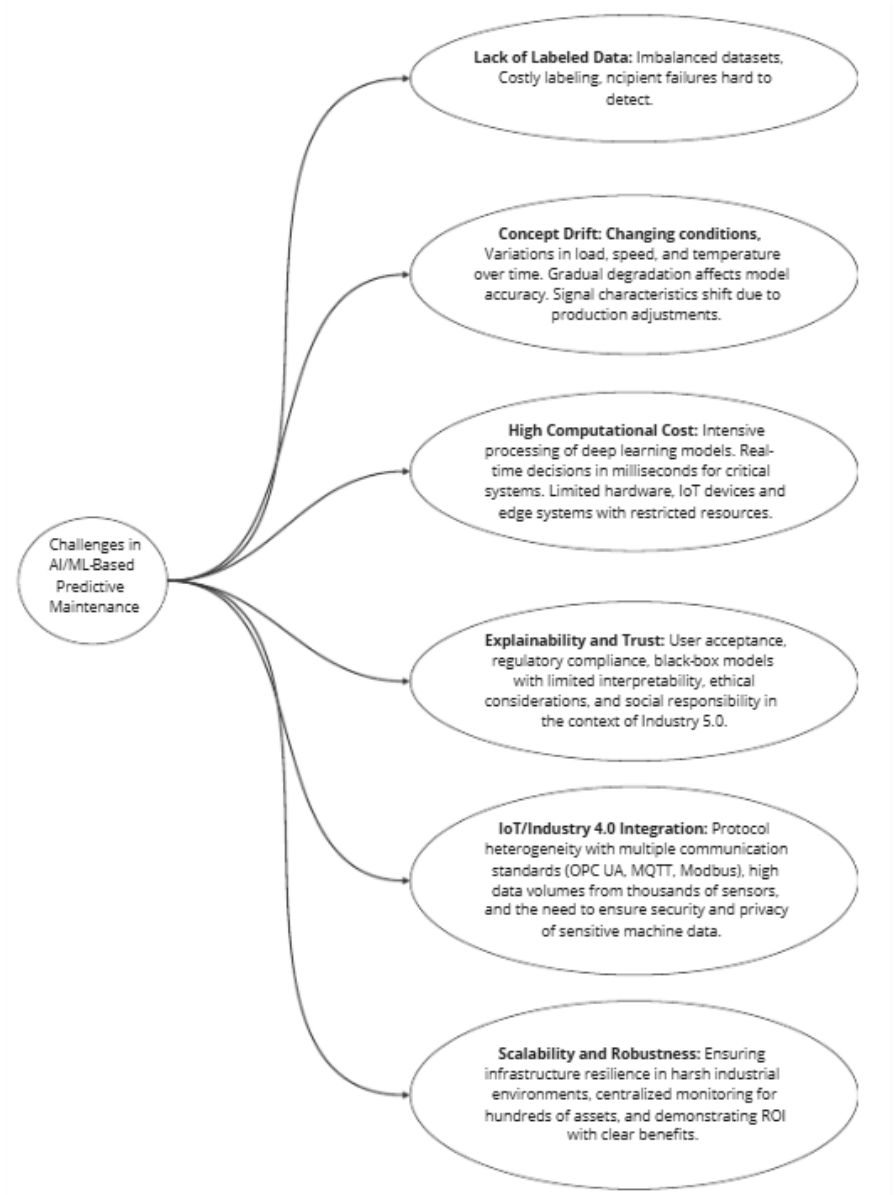


Figure 4: Key Challenges in AI/ML-Based Predictive Maintenance [20]

Organisational readiness also plays a role. The introduction of an ML-supported maintenance workflow demands user trust, clear visualisation, and gradual adoption. Maintenance teams must be able to understand the model's suggestions and relate them to physical behaviours on the line. Early adoption would therefore require a hybrid approach where human expertise validates the model's outputs, enabling the system to adapt and mature over time.

Despite these limitations, the framework lays the foundation for a scalable predictive maintenance ecosystem. It creates a pathway for future enhancements such as real-time anomaly detection, sensor-level diagnostics, and eventually AI agents capable of retrieving relevant past work orders, summarising them automatically, and supporting technicians with recommendations. As interoperability increases between CMMS, monitoring platforms, and analytics tools, the framework can evolve from a structured prototype into a fully integrated decision-support system.

While the proposed ML architecture is intentionally conservative and aligned with current data constraints, it represents a meaningful step toward transitioning Scania's battery production maintenance from reactive and manual practices to a more proactive, data-driven, and eventually autonomous approach.

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